GROUP 7

Revised Healthcare Big Data Case Studies

Huatuo-26M Data Integration, Hadoop MapReduce, and Apache Spark

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Revision Rationale

Huatuo-26M integrates multiple heterogeneous medical question-answer sources. The full consultation source provides questions and answer URLs, while the encyclopedia and knowledge-graph sources provide questions and textual answers. Huatuo26M-Lite additionally provides an ID, answer-quality score, department label, and related disease. The published datasets do not provide hospital organization, doctor name, or timestamp fields.

The case studies below are revised to use fields that genuinely exist. Where disease or department values are added through dictionary matching, they are explicitly marked as inferred rather than presented as original dataset metadata.

Unified Dataset Fields

- **Core observed fields:** record_id, source, question, answer, answer_type.
- **Lite observed fields:** quality_score, department, related_disease.
- **Derived fields:** question_length, answer_length, question_hash, duplicate_flag, question_type.
- **Enriched fields:** inferred_department, inferred_disease, and metadata_origin.

CASE STUDY 1

Medical Department Demand and Resource Allocation Analysis

Explanation

This case study analyzes the distribution of medical questions across departments. Observed department labels from Huatuo26M-Lite are counted directly. For records without a department, an optional enrichment stage matches disease and medical keyword dictionaries to infer a likely department. Observed and inferred results are reported separately.

Aim

To identify high-demand medical departments and demonstrate how distributed aggregation can support healthcare resource-allocation analysis.

Input

- record_id, source, question
- department and related_disease, when observed in Huatuo26M-Lite
- inferred_department and metadata_origin, when enrichment is enabled

Techniques / Functions

Filtering, dictionary-based pattern matching, mapping, grouping, counting, sorting, and top-N retrieval.

MapReduce Design

Mapper: emit (department, metadata_origin), 1

Reducer: sum counts for each department and origin

Output

A ranked department-demand report containing consultation counts, percentages, and metadata origin. The report will clearly separate observed labels from inferred labels.

Hadoop and Spark Equivalence

Hadoop Streaming and PySpark will apply the same filtering, department dictionary, grouping key, and ranking rules to produce comparable outputs.

CASE STUDY 2

Disease and Symptom Trend Mining

Explanation

This case study identifies frequently mentioned diseases and symptoms across the unified medical QA dataset. A disease dictionary is built from observed related-disease values in Huatuo26M-Lite. Disease and symptom terms are then matched against questions and textual answers from all sources.

Aim

To discover common medical concerns, disease patterns, and symptom associations across large-scale heterogeneous medical text.

Input

- source, question, answer, answer_type
- observed related_disease values and curated symptom dictionary

Techniques / Functions

Text normalization, WordCount, pattern matching, filtering, grouping, counting, co-occurrence calculation, sorting, and top-N retrieval.

MapReduce Design

Mapper: emit matched disease or symptom, 1

Reducer: sum mentions and rank terms by frequency

Output

A ranked frequency report of diseases and symptoms, plus the most frequent disease-symptom co-occurrence pairs. Results may also be grouped by source to show differences between consultation, encyclopedia, knowledge-graph, and Lite records.

Hadoop and Spark Equivalence

Both frameworks will use identical normalization rules and dictionaries. Textual answers will be analyzed, while URL-only answers will be excluded from answer-text matching and counted separately.

CASE STUDY 3

Medical QA Quality and Completeness Analysis

Explanation

This case study evaluates data quality and completeness across the integrated Huatuo sources. It replaces the unsupported hospital-performance analysis because hospital organization and doctor-name fields are not publicly available. The analysis measures missing values, URL-only answers, duplicates, question and answer lengths, and the quality-score distribution available in Huatuo26M-Lite.

Aim

To compare source-level data quality, identify records requiring cleaning, and evaluate the effect of preprocessing on the unified dataset.

Input

- source, question, answer, answer_type
- quality_score and department, when available in Huatuo26M-Lite
- question_length, answer_length, question_hash, duplicate_flag

Techniques / Functions

Partitioning by source, filtering, duplicate detection, counting, average/mean calculation, min/max retrieval, percentage calculation, grouping, and sorting.

MapReduce Design

Mapper: emit source and metric, partial statistics

Reducer: aggregate counts, sums, minima, and maxima

Output

A source-comparison report showing record counts, missing-field rates, duplicate rates, URL-answer percentages, average text lengths, and Lite quality-score statistics. It will also compare dataset quality before and after cleaning.

CASE STUDY 4

Medical Question-Type Distribution and Framework Performance Analysis

Explanation

This case study classifies medical questions into intent categories and compares Hadoop and Spark processing performance. It replaces temporal analysis because the published datasets do not provide timestamps. Example categories include symptoms, causes, diagnosis, treatment, medication, prevention, and other.

Aim

To identify the most common types of medical information sought and evaluate the processing-time scalability of Hadoop and Spark using the same classification and aggregation task.

Input

- record_id, source, question
- shared question-type pattern dictionary

Techniques / Functions

Text normalization, pattern matching, classification, partitioning, grouping, aggregation/counting, sorting, and processing-time calculation.

MapReduce Design

Mapper: classify question and emit question_type, 1

Reducer: sum questions for each question type

Output

A ranked question-type distribution report and a Hadoop-versus-Spark performance report. Performance will be measured using equivalent jobs over multiple input sizes, with repeated runs and average execution times.

Performance Evaluation Method

- Execute equivalent Hadoop Streaming and PySpark jobs.
- Test the same 25%, 50%, 75%, and 100% input partitions.
- Run each experiment at least three times and calculate the average.
- Record environment, worker count, partition count, processing time, and output count.

Data Processing and Integration Scope

The unified dataset will be created by schema standardization and unioning records from the Huatuo sources. The sources will not be joined as though they describe the same consultations because no reliable shared consultation identifier exists.

Preprocessing Lifecycle

- **Data aggregation:** collect the consultation, encyclopedia, knowledge-graph, and Lite sources.
- **Data integration:** map all source schemas into one canonical JSON Lines schema and union the records.
- **Data cleaning:** normalize arrays and whitespace, remove empty records, validate URLs, and mark duplicates.
- **Data transformation:** derive lengths, hashes, answer types, question types, and metadata-origin fields.
- **Data enrichment:** optionally infer missing diseases and departments using transparent dictionaries derived from observed Lite metadata.

Fair Framework Comparison

Every Hadoop and Spark implementation will receive the same standardized HDFS input and apply the same business rules. Downloading and one-time preprocessing will be reported separately from analytical job execution time. Output record counts and aggregate totals will be validated to confirm that both frameworks produce equivalent results.

Dataset Integrity Statement

No hospital organizations, doctor names, timestamps, or synthetic quality scores will be invented. Any inferred metadata will include its origin and will be analyzed separately from observed metadata. This maintains traceability and prevents unsupported conclusions.

Expected Contribution

The revised case studies demonstrate distributed filtering, mapping, grouping, counting, sorting, retrieval, pattern matching, partitioning, average calculation, min/max retrieval, classification, and performance measurement while remaining aligned with the actual Huatuo-26M data.